

Tide retrospective: *partition derivatives are unperturbed Gibbs moments*

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Abstract

A short capstone to the anharmonic partition-derivative arc. For the 1D anharmonic potential, the n -th h -derivative of the perturbed partition Z at $h = 0$, normalised by $Z(0)$, is the unperturbed Gibbs expectation of the observable $-(tx)^n$:

$$\frac{\text{iteratedDeriv } n \, Z \, (0)}{Z(0)} = \text{gibbsExp } \mu \, L \, \text{id } t \, 0 \, (x \mapsto -(tx)^n).$$

This ties the arc to the `Threepoint.gibbsExp` machinery used by the FDT capstone. The proof is a six-line definitional unfolding on top of the arc's `iteratedDeriv` identity. New file `AnharmonicMomentNormalisation.lean`; builds clean, no `sorry/axiom/native.decide`.

1 Setting

This session built a three-tide arc on the laplace seabed: the anharmonic FDT/cross-susceptibility identities made unconditional¹, the general- n iterated-derivative step², and its general- h strengthening with formal $Z''(0)$, $Z'''(0)$ ³. Those produce the derivatives $\text{iteratedDeriv } n \, Z \, (0) = \int (-(tx))^n e^{-tL}$ as bare integrals. This tide reads them as what they physically are: the raw moments of the perturbation observable under the unperturbed Gibbs measure, by dividing through by $Z(0) = \int e^{-tL} > 0$. It was the lowest-race candidate available (laplace quiet, the whole arc warm in cache) and closes the arc's narrative by connecting it to the `gibbsExp` posterior-expectation definition that the FDT capstone consumes.

2 The thing formalised

For the anharmonic potential, $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$, and every $n \in \mathbb{N}$:

$$\frac{\text{iteratedDeriv } n \, Z \, (0)}{Z(0)} = \langle (-(tx))^n \rangle_0,$$

where $Z = \text{weightedPartition} \dots 0$ and $\langle \cdot \rangle_0$ is the unperturbed ($h = 0$) Gibbs expectation.

¹2026-05-30-tide-anharmonic-fdt-capstone.

²2026-05-30-tide-anharmonic-partition-higher-deriv.

³2026-05-30-tide-anharmonic-partition-deriv-general-h.

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theorem iteratedDeriv_partition_div_eq_gibbsExp (n : ℕ) ... :
  iteratedDeriv n (weightedPartition lam alpha gamma t 0) 0
    / weightedPartition lam alpha gamma t 0 0
  = Threepoint.gibbsExp volume (anharmonicPotential lam alpha gamma)
    (fun x => x) t 0 (fun x => (-(t * x)) ^ n)

```

3 Proof strategy

Definitional. Rewrite the iterated derivative by the arc’s identity $\text{iteratedDeriv } n \, Z(0) = \text{weightedPartition } \dots \, n \, 0 = \int (-(tx))^n e^{-t(L+0 \cdot x)}$. Unfold `weightedPartition` and `gibbsExp`. The two sides are then quotients with identical numerators $\int (-(tx))^n e^{-t(L+0 \cdot x)}$; the only difference is the denominator, $\int (-(tx))^0 e^{\dots}$ on the left versus $\int e^{\dots}$ on the right. `simp only [pow_zero, one_mul]` collapses $(-(tx))^0 = 1$ (and β -reduces the `gibbsExp` observable application), closing the goal by reflexivity.

4 Roadblocks and resolutions

One: the first attempt peeled the division with `congr 1` and then tried `funext`, but `congr` on $A/B = C/D$ leaves an *integral* equality $\int \dots = \int \dots$ (a real-number equation), not a function equality, so `funext` could not unify. Replacing the `congr/funext` dance with a single `simp only [pow_zero, one_mul]` both reduced the denominator and β -reduced the `gibbsExp` numerator, closing the goal directly.

5 What was Mathlib, what was new

Mathlib. `pow_zero`, `one_mul`, and `simp`’s β -reduction. **Seabed (reused).** The arc’s `iteratedDeriv` identity⁴; `Threepoint.gibbsExp`. **New.** The normalisation bridge itself — about six lines of proof.

6 Lessons

- **congr on a quotient leaves integral equalities, not function equalities.** When the remaining goal is $\int f = \int g$, reach for `simp/integral_congr`, not `funext`.
- **A definitional bridge is a legitimate (small) tide.** Connecting the arc’s bare-integral derivatives to the `gibbsExp` posterior expectation is what makes “ $Z^{(n)}(0)$ ” mean “ n -th moment” to a reader of the primer; the value is in the identification, not the line count.

7 Follow-ups

- **Fourth-cumulant / flow-equation identity.** With the moments now named (and $Z^{(4)}(0)$ a further instance), assemble $\kappa_4 = \partial_h^4 \log Z|_0$ — the arc’s true payoff.
- **Promote `amgm_t_abs_x` to `lean-common`.** Load-bearing across the whole arc now.

⁴`iteratedDeriv_weightedPartition_zero` and `weightedPartition`, both from this session.